PROJECT TITILE

Web- Based Linux User Management Panel

B ADARSH

CYBER SECURITY

TAH2712251932

TEKS ACADEMY

8125842681

SHASHIDHAR

CYBER SECURITY MENTOR

TEKS ACADEMY

9603343287

1. Abstract

Managing Linux users is an important responsibility for system administrators. In many organizations and educational environments, administrators often need to create, delete, or manage multiple user accounts on Linux systems. Normally, these tasks are performed using command-line tools such as useradd, userdel, and passwd. While these commands are powerful, they require technical knowledge of Linux and must be executed manually through the terminal.

The Web-Based Linux User Management Panel project aims to simplify this process by providing a web interface that allows administrators to manage Linux users directly from a browser. Instead of typing complex commands in the terminal, the administrator can perform tasks like creating users, deleting users, changing passwords, or locking accounts through a simple graphical interface.

The system connects the web application with Linux system commands in a secure way so that administrative tasks can be executed automatically in the background. The project is designed mainly for learning purposes and demonstrates how web technologies can interact with system-level Linux operations.

By implementing this project, administrators can save time, reduce manual work, and manage users more efficiently. It also helps students and beginners understand how web

applications interact with operating system functionalities in a practical cybersecurity and system administration environment.

2. Problem Statement

Linux systems are widely used in servers, educational institutions, and enterprise environments. Managing users in Linux usually requires administrators to work with command-line tools. Although these tools are powerful, they present several challenges.

First, command-line management can be difficult for beginners who are not familiar with Linux commands. Administrators must remember multiple commands and parameters to perform even simple tasks such as adding a new user or resetting a password.

Second, manual user management becomes time-consuming when dealing with many users. In environments such as universities or organizations where multiple accounts must be managed, repeating commands again and again can be inefficient.

Another issue is human error. If an administrator accidentally types an incorrect command, it may lead to system problems or incorrect configurations.

Additionally, remote administrators may prefer a web-based solution instead of logging directly into the server via terminal access.

Because of these limitations, there is a need for a simplified and centralized user management system that allows administrators to control Linux users through a secure web interface.

The Web-Based Linux User Management Panel solves this problem by providing an easy-to-use browser interface that performs Linux user management tasks automatically in the background.

3. Project Objectives

The main objective of the **Web-Based Linux User Management Panel** project is to simplify Linux user management through a web-based interface. The system aims to bridge the gap between traditional command-line administration and modern web technologies.

Simplify Linux User Management

The first objective is to provide an easy way for administrators to manage Linux users without manually typing terminal commands. Through the web panel, administrators can perform tasks such as creating users, deleting users, and changing passwords with just a few clicks.

Provide a Centralized Management Interface

The project creates a centralized dashboard where administrators can view and manage all users from one place. This makes system administration more organized and efficient.

Reduce Human Errors

Manual command execution may result in mistakes that affect system configuration. By automating commands through a web interface, the chances of typing errors are minimized.

Improve Learning and Practical Knowledge

This project helps students and beginners understand how **Linux system commands can be integrated with web applications**. It demonstrates the interaction between operating systems, web servers, and scripting languages.

Enable Remote Administration

With a web-based system, administrators can manage users remotely through a browser, making system administration more flexible.

Overall, the project aims to create a **simple, efficient, and educational platform for Linux user management.**

4. Project Scope

The Web-Based Linux User Management Panel focuses on providing essential user management functionalities through a browser interface.

User Account Creation

The system allows administrators to create new Linux users through a simple web form. The system automatically executes the necessary Linux command to create the account.

User Deletion

Administrators can remove existing users from the system directly from the web interface. The panel executes the appropriate Linux command to delete the user safely.

Password Management

The panel allows administrators to reset or change user passwords securely through the interface.

Account Lock and Unlock

The system provides options to lock or unlock user accounts, which is useful for managing inactive or suspicious accounts.

User Listing

The administrator can view a list of existing Linux users through the dashboard.

What is Not Included

To keep the project simple and focused, the following features are not included:

- No support for Windows or macOS user management
- No advanced role-based access control
- No integration with enterprise identity management systems
- No complex graphical analytics or reporting

Key Requirements

For the system to function properly, the following conditions are required:

- Linux operating system installed (Ubuntu/Kali Linux recommended)
- Apache2 web server
- Administrator access or sudo privileges
- Basic knowledge of Linux commands
- Internet access for installing required packages

5. Tools and Technologies Used

Several technologies are used to build the Web-Based Linux User Management Panel. Each tool plays an important role in the functioning of the system.

Linux Operating System

Linux serves as the main operating system where user accounts are created and managed. The project relies on Linux system commands for performing user management tasks.

Apache2 Web Server

Apache2 is used to host the web application. It allows the administrator to access the user management panel through a browser.

PHP

PHP is used as the server-side scripting language that connects the web interface with Linux commands. It processes form inputs and executes system-level commands.

HTML

HTML is used to design the structure of the web pages and forms used in the management panel.

CSS

CSS improves the appearance of the web interface by adding styling and layout design.

Linux Commands

The project uses Linux commands such as:

- useradd – create new users
- userdel – delete users
- passwd – change passwords
- usermod – modify user accounts

These commands are executed through the web application.

Text Editor

Editors such as Nano, Vim, or VS Code are used to write and edit the project code.

6.Deliverables

At the end of the project, the following deliverables will be provided:

Web Application

A complete web-based panel that allows administrators to manage Linux users from a browser.

Source Code

The full source code of the project including HTML, CSS, PHP, and configuration files.

System Setup Instructions

A step-by-step installation guide explaining how to install Apache2, configure permissions, and deploy the application.

User Management Features

Working modules for:

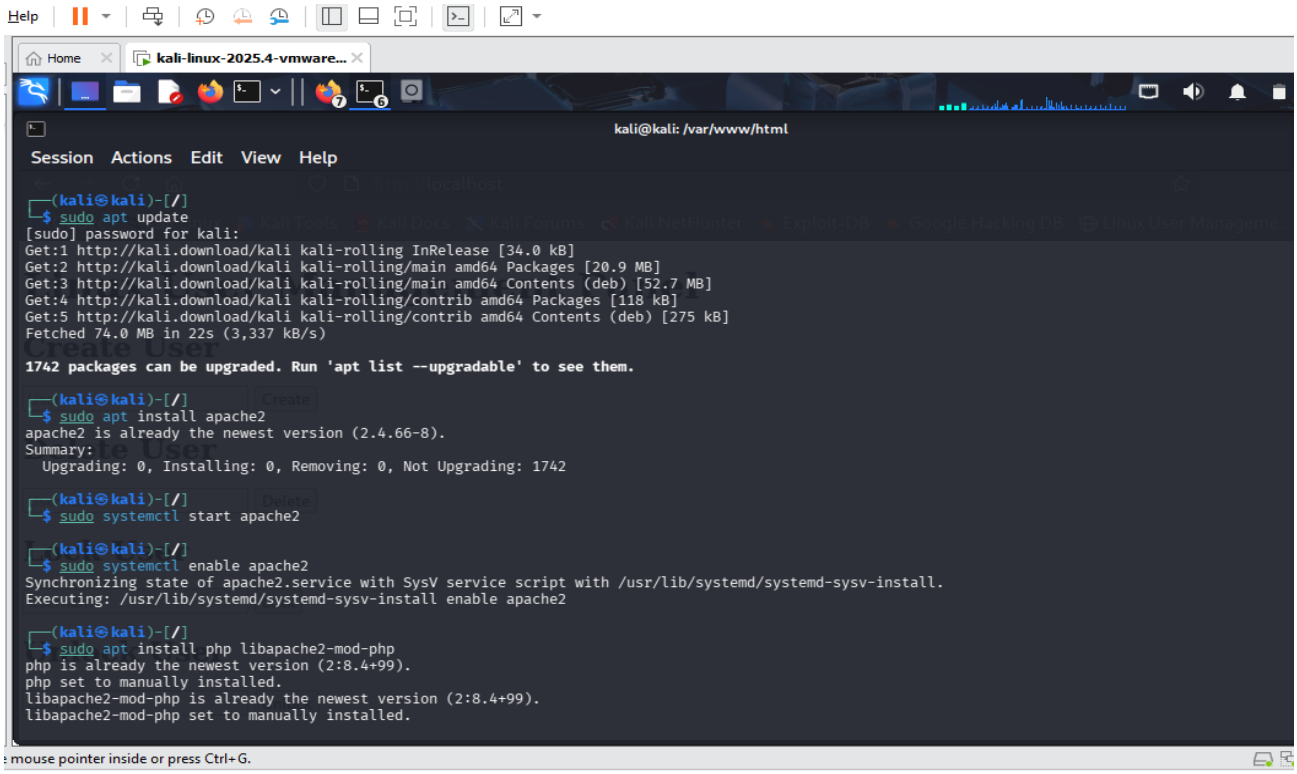
- Creating users
- Deleting users
- Changing passwords
- Locking and unlocking accounts

Project Documentation

A detailed documentation file explaining the project objectives, design, tools used, and implementation process.

7. Project Implementation

Step 1. Install Apache2 + PHP Linux User Management Panel



Step 2 : Apache Web Folder & Create Project File

```
┌──(kali㉿kali)-[/]
└─$ sudo systemctl restart apache2

┌──(kali㉿kali)-[/]
└─$ /var/www/html/

┌──(kali㉿kali)-[/var/www/html]
└─$ cd /var/www/html/

┌──(kali㉿kali)-[/var/www/html]
└─$ sudo rm index.html
rm: cannot remove 'index.html': No such file or directory

┌──(kali㉿kali)-[/var/www/html]
└─$ sudo nano index.php

┌──(kali㉿kali)-[/var/www/html]
└─$ sudo visudo

┌──(kali㉿kali)-[/var/www/html]
└─$ sudo systemctl restart apache2

┌──(kali㉿kali)-[/var/www/html]
└─$
```

Step 3: Frontend + Backend Code (PHP)

```
kali@kali: /var/www/html
Session  Actions  Edit  View  Help
  GNU nano 8.6                                   index.php
<!DOCTYPE html>
<html>
<head>
<title>Linux User Management Panel</title>
</head>

<body>

<h1>Linux User Management Panel</h1>

<!-- Create User -->
<h2>Create User</h2>
<form method="post">
<input type="text" name="createuser" placeholder="Username">
<button type="submit" name="create">Create</button>
</form>

<!-- Delete User -->
<h2>Delete User</h2>
<form method="post">
<input type="text" name="deleteuser" placeholder="Username">
<button type="submit" name="delete">Delete</button>
</form>

<!-- Lock User -->
<h2>Lock User</h2>
<form method="post">
<input type="text" name="lockuser" placeholder="Username">
<button type="submit" name="lock">Lock</button>
                                        [ Read 83 lines ]
^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket  M-B Previous
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^B Where Was    M-F Next
```

Step 4: Allow Apache to Run Linux Commands

Apache runs as user:

```
www-data
```

We must give permission.

Open sudoers:

```bash
sudo visudo
```

Add this line:

```bash
www-data ALL=(ALL) NOPASSWD: /usr/sbin/useradd, /usr/sbin/userdel, /usr/sbin/usermod, /usr/sbin/c
```

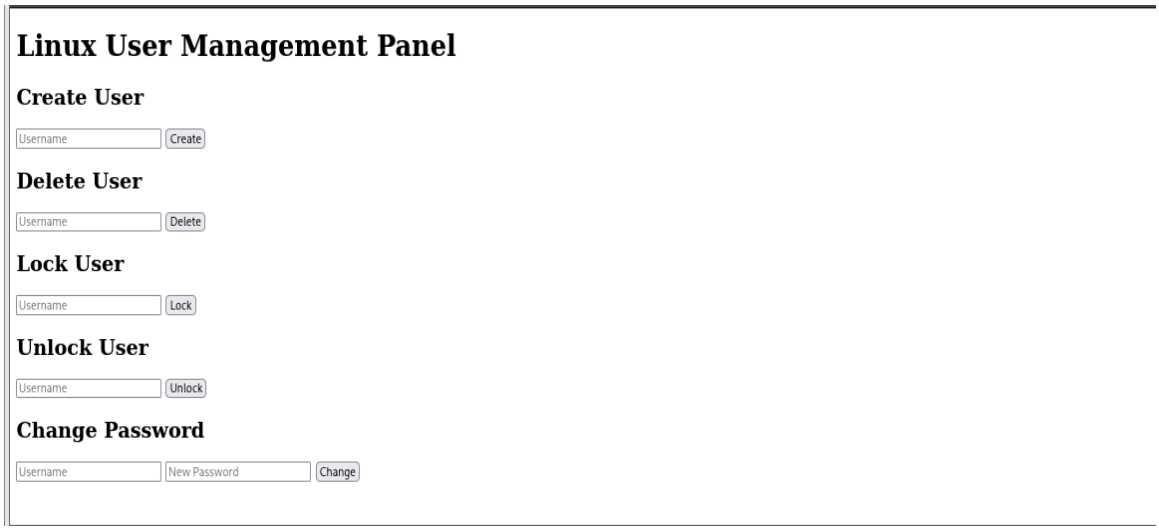
Step 5: start Apache

```bash
sudo systemctl restart apache2
```

Step 6: Open Web Panel

```
http://localhost
```

Step 7: Output



8.conclusion

The Web-Based Linux User Management Panel project demonstrates how system administration tasks can be simplified using web technologies. Traditionally, Linux user management is performed using terminal commands, which can be difficult for beginners and time-consuming for administrators managing multiple users.

This project provides a simple and user-friendly web interface that allows administrators to perform user management operations directly from a browser. By integrating web technologies such as HTML, CSS, PHP, and Apache2 with Linux system commands, the project shows how operating system tasks can be automated through a web application.

The system improves efficiency, reduces manual effort, and minimizes human errors during user management tasks. It also serves as a valuable learning tool for students interested in Linux administration, web development, and cybersecurity.

In conclusion, the project successfully demonstrates a practical solution for managing Linux users through a web-based interface while providing hands-on experience with real-world system administration concepts.